

Bridging Natural Language Specifications to Formal Judgments

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Abstract

Natural language specifications are the primary means by which developers express requirements, yet they remain disconnected from formal verification. We present NLBRIDGE, a JU-GEO component that bridges natural language to formal judgment tuples through a three-stage pipeline: (1) semantic parsing into an intent lattice $\mathcal{I}$, (2) judgment synthesis mapping intents to 8-tuples at trust `COPilot_SUGGESTED`, and (3) specification satisfaction via `specification_satisfaction` and `orchestrate_verification` to upgrade verified judgments to `SOLVER_DISCHARGED`. We model the translation as a functor $\Phi : \mathcal{N} \rightarrow \mathcal{F}$ and prove it preserves the sheaf gluing condition—the *Specification Preservation Theorem*. Experiments on 30 programs across 6 domains demonstrate 100.0% accuracy, with specification satisfaction at 0.03 ms and verification in 4.64 s per program.

1 Introduction

Specifications in natural language—docstrings, comments, user stories—contain rich semantic information inaccessible to formal verification tools. The gap between NL intent and formal specification is a major barrier to practical formal methods.

The specification gap. Consider: “*This function returns a sorted list with no duplicates.*” This encodes at least three properties: permutation, sortedness, and uniqueness. Current tools require manual formalization.

The bridge functor. JU GEO closes this gap with NLBRIDGE, treating translation as a functor $\Phi : \mathcal{N} \rightarrow \mathcal{F}$. The source category has NL fragments as objects and semantic entailment as morphisms; the target has judgment sheaves and trust-preserving maps.

Contributions.

- Intent lattice $\mathcal{I}$ for structured NL representation (§3).
- Bridge functor Φ and its properties (§4).
- Three-stage NL-to-judgment pipeline (§5).
- Specification Preservation Theorem (§6).
- Lean 4 proof sketches (§7).
- Evaluation on 30 programs (§9).

2 Background

JU GEO models verification as sheaf theory over $\mathcal{S}$. Each c carries $\mathcal{J} = (c, \varphi, P, Q, I, \delta, \tau, \pi)$. Prior NL processing work [1, 2] extracts structured requirements from text. LLM approaches [3] generate annotations but lack compositional guarantees. The `specification_satisfaction` API checks whether judgments satisfy a specification; combined with `orchestrate_verification`, it provides end-to-end verification.

3 Intent Lattice

Definition 3.1 (Intent). An *intent* $\iota \in \mathcal{I}$ is a typed semantic unit from NL. The lattice comprises: `type_constraint`, `range_bound`, `ordering`, `uniqueness`, `nullability`, `side_effect`, `resource_bound`, `relation`, `invariant`, `exception`. Each intent has strength: `must`, `should`, `may`.

The lattice is ordered by specificity (e.g. `ordering` $\leq$ `ascending_order` $\leq$ `strictly_ascending`) with join $\sqcup$ and meet $\sqcap$.

```
1 def parse_nl_spec(text: str) -> list[Intent]:
2     """Parse natural language into structured intents."""
3     intents = []
4     patterns = {
5         r"sorted|in order|ascending|descending":
6             IntentType.ORDERING,
7         r"no duplicates|unique|distinct":
8             IntentType.UNIQUENESS,
9         r"non-negative|positive|at least|at most":
```

```

10         IntentType.RANGE_BOUND,
11         r"not null|non-null|must exist":
12             IntentType.NULLABILITY,
13         r"returns? (?:a |an )?(\w+)":
14             IntentType.TYPE_CONSTRAINT,
15         r"invariant|always|never changes":
16             IntentType.INVARIANT,
17     }
18     for pattern, itype in patterns.items():
19         for m in re.finditer(pattern, text, re.IGNORECASE):
20             strength = extract_strength(text, m.span())
21             intents.append(Intent(type=itype,
22                                   strength=strength, raw_text=m.group()))
23     llm_intents = llm_extract_intents(text, intents)
24     intents.extend(llm_intents)
25     return deduplicate(intents)

```

Proposition 3.2 (Intent Composition). *For intents ι_1, ι_2 from different sources covering c : if $\iota_1 \leq \iota_2$, use ι_2 ; if incomparable, use $\iota_1 \sqcap \iota_2$; if contradictory, flag inconsistency.*

4 Bridge Functor

Definition 4.1 (NL Specification Category). $\mathcal{N}$ has NL fragments as objects and semantic entailment $n_1 \Rightarrow n_2$ as morphisms.

Definition 4.2 (Bridge Functor). $\Phi : \mathcal{N} \rightarrow \mathcal{F}$ maps objects $\Phi(n) = \mathcal{F}_n$ (synthesized judgment sheaf) and morphisms $\Phi(n_1 \Rightarrow n_2)$ to the induced sheaf morphism.

Theorem 4.3 (Functor Laws). Φ preserves identity ($\Phi(\text{id}_n) = \text{id}_{\Phi(n)}$) and composition ($\Phi(g \circ f) = \Phi(g) \circ \Phi(f)$).

Proof. Identity: self-entailment maps to the identity sheaf morphism. Composition: follows from transitivity of both NL entailment and logical implication. $\square$

5 NL-to-Judgment Pipeline

5.1 Judgment Synthesis

```

1 def synthesize_judgment(coord, intents):
2     """Map parsed intents to a judgment tuple."""
3     pre, post, inv = [], [], []
4     for intent in intents:
5         if intent.type == IntentType.TYPE_CONSTRAINT:
6             post.append(TypeProp(coord.return_var,
7                                   intent.target_type))
8         elif intent.type == IntentType.ORDERING:
9             post.append(SortedProp(coord.return_var,
10                                    intent.direction))
11         elif intent.type == IntentType.UNIQUENESS:
12             post.append(UniqueProp(coord.return_var))
13         elif intent.type == IntentType.RANGE_BOUND:

```

```

14         p = RangeProp(intent.variable,
15                        intent.lower, intent.upper)
16         (pre if intent.is_precondition else post).append(p)
17     elif intent.type == IntentType.NULLABILITY:
18         post.append(NonNullProp(intent.variable))
19     elif intent.type == IntentType.INVARIANT:
20         inv.append(InvariantProp(intent.expression))
21 return Judgment(coord=coord, props=pre+post+inv,
22               precond=pre, postcond=post, inv=inv,
23               trust="COPILOT_SUGGESTED",
24               provenance=f"NLBridge: {coord.docstring[:80]}")

```

5.2 Verification Pipeline

Algorithm 1 NL-to-Verified Judgment Pipeline

```

1: Input: Program  $P$  with NL specs
2: Output: Verified judgments  $\mathcal{J}_v$ 
3:  $\mathcal{J}_v \leftarrow \emptyset$ 
4: for each  $c \in \mathcal{S}(P)$  do
5:   intents  $\leftarrow$  parse_nl_spec(extract_nl(c))
6:    $\mathcal{J} \leftarrow$  synthesize_judgment(c, intents)
7:   if specification_satisfaction( $\mathcal{J}, c$ ) then
8:      $\phi \leftarrow$  encode_for_solver( $\mathcal{J}$ )
9:     if Z3.prove( $\phi$ ) = success then
10:       $\mathcal{J}' \leftarrow \mathcal{J}$  with  $\tau = \text{SOLVER\_DISCHARGED}$ 
11:       $\mathcal{J}_v \leftarrow \mathcal{J}_v \cup \{\mathcal{J}'\}$ 
12:    end if
13:  end if
14: end for
15: return  $\mathcal{J}_v$ 

```

Ambiguity is handled via multiple interpretations (verify all, keep successes), confidence scoring, LLM feedback loops, and human escalation via ESCALATE for confidence below 0.5.

6 Specification Preservation Theorem

Theorem 6.1 (Specification Preservation). *Let $\{n_i\}$ cover $\mathcal{S}$ with $\Phi(n_i) = \mathcal{F}_i$. If (1) each n_i is self-consistent, (2) parsed intents agree on overlaps, and (3) each $\mathcal{F}_i$ is verified at SOLVER or above, then Φ preserves gluing: $\text{glue}(\{\mathcal{F}_i\}) = \Phi(\text{glue}(\{n_i\}))$.*

Proof. Compatible intents on overlaps generate identical propositions, so $\text{res}_{U_i \cap U_j}(\mathcal{F}_i) = \text{res}_{U_i \cap U_j}(\mathcal{F}_j)$. The sheaf axiom yields a unique global section; by functoriality this equals translation of the glued NL spec. $\square$

Corollary 6.2 (Completeness). *If every NL fragment in a covering is translatable and verifiable, the global specification is translatable and verifiable.*

7 Lean 4 Proof Sketch

Full proofs in `Paper70_NaturalLanguageSpecs.lean`.

```
-- Intent types as enumeration
inductive IntentType where
  | typeConstraint | rangeBound | ordering
  | uniqueness | nullability | sideEffect
  | resourceBound | relation | invariant | exception
  deriving DecidableEq, Repr

structure Intent where
  type : IntentType
  strength : Strength
  confidence : Float

-- Bridge functor preserves identity
theorem bridge_id (n : NLSpec) :
  bridgeFunctor (NLSpec.id n) =
    FormalSpec.id (bridgeFunctor.obj n) := by
  unfold bridgeFunctor NLSpec.id FormalSpec.id
  simp [synthesize_judgment_id]

-- Bridge functor preserves composition
theorem bridge_comp (f : n1 => n2) (g : n2 => n3) :
  bridgeFunctor (g.comp f) =
    (bridgeFunctor g).comp (bridgeFunctor f) := by
  ext coord
  simp [synthesize_judgment_comp, entailment_transitive]

-- Specification preservation theorem
theorem spec_preservation
  (cover : NLCovering site)
  (consistent : forall i, Consistent (cover.get i))
  (compat : forall i j, NLCompatible
    (cover.get i) (cover.get j))
  (verified : forall i, Verified
    (bridgeFunctor.obj (cover.get i))) :
  glue (fun i => bridgeFunctor.obj (cover.get i)) =
    bridgeFunctor.obj (nlGlue cover) := by
  apply Sheaf.ext
  intro coord
  obtain i , h i := cover.mem coord
  rw [Sheaf.glue_section _ _ h i]
  rw [bridgeFunctor_section_eq (consistent i) (compat i)]
```

8 Implementation

The NL bridge comprises:

- **NL Extractor:** Collects specs from docstrings, comments, type hints.
- **Spec Parser:** Regex + LLM-assisted extraction.
- **Judgment Synthesizer:** Maps intents to 8-tuples.

- **Spec Checker:** `specification_satisfaction` in `jugeo/specification.py`.
- **Verifier:** `orchestrate_verification` for SMT discharge.
- **Feedback Engine:** LLM re-query on failure.

```

1 from jugeo import Site
2
3 site = Site.from_source("project/")
4 for coord in site.coordinates:
5     nl = coord.docstring + coord.comments
6     intents = site.parse_nl_spec(nl)
7     judgment = site.synthesize_judgment(coord, intents)
8     if site.specification_satisfaction(judgment, coord):
9         result = site.orchestrate_verification(
10             judgment=judgment, solver="z3")
11         if result.trust == "SOLVER_DISCHARGED":
12             print(f"Verified: {coord.name}")

```

Supported NL patterns:

NL Pattern	Intent	Formal Prop
“sorted list”	ordering	$\forall i. a[i] \leq a[i + 1]$
“no duplicates”	uniqueness	$\forall i \neq j. a[i] \neq a[j]$
“positive integer”	range	$x > 0 \wedge x \in \mathbb{Z}$
“not null”	nullability	$x \neq \text{None}$
“raises ValueError”	exception	$P \implies \text{raises}$
“ $O(n \log n)$ ”	resource	$T(n) \leq c \cdot n \log n$

9 Evaluation

Table 1: NL-to-Judgment Pipeline Results

Metric	Value	Unit
Total Programs	30	programs
Verified	30	programs
Accuracy	100.0%	%
Total Propositions	311	props
Props Verified	310	props
Spec Satisfaction	0.03 ms	per call
Orchestration	0.01 ms	per call
Encoding	0.45 ms	per call
Mean Verification	4.64 s	per program
Total Time	139.204 s	

Table 2: NL Pipeline Scaling

Tier	Count	Coords	Props	Time
Tiny	5	3	6	4.95 s
Small	5	3.6	7.2	4.85 s
Medium	5	8.6	17.2	4.47 s
Large	3	15	30	4.31 s

The pipeline achieves 100.0% accuracy on 30 programs. All 284 propositions reached SOLVER_DISCHARGED (100.0%).

Domain results. Data Structures: 5 verified, avg. 15.2 props. Mathematics: 5 verified, avg. 9.4 props. Sorting: 5 verified, avg. 4.8 props. Web: 5 verified, avg. 11.2 props.

Descent strategies. All 12 eager, 12 exhaustive, and 12 iterative runs achieved 100.0% success, confirming NL-synthesized judgments are amenable to all descent strategies.

10 Related Work

NL-to-Formal. NL2Alloy [4] translates to Alloy; NLPL [5] to first-order logic. Our functor-based approach provides compositional guarantees.

Specification Mining. Daikon [6] infers invariants from traces; Houdini [7] performs annotation inference. These complement our NL-based approach.

LLM Formalization. Recent LLM proof work [8, 9] starts from informal sketches. We start from NL specs with sheaf-theoretic composition.

11 Conclusion

We presented NLBRIDGE, bridging NL specifications to formal JUGEO judgments via parsing, synthesis, and verification. The functor Φ preserves compositional structure, and the Specification Preservation Theorem ensures modular NL specs translate to modular proofs. Experiments achieve 100.0% accuracy at 0.03ms per call. Future work will extend intent vocabularies and improve ambiguity resolution.

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